A Anchor calibration

Yuan 2025 GCST90624411 · GTEx LAA

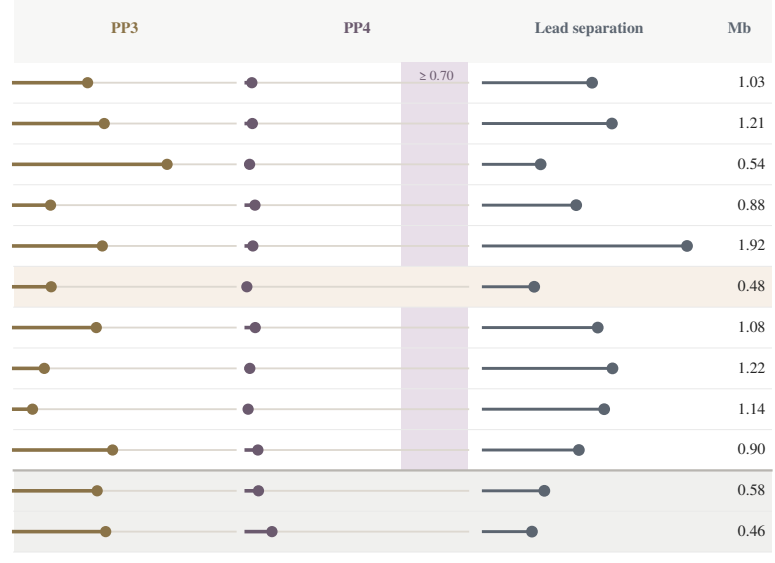
Family / role	Anchor	Local GWAS 7/12	Atrial eGene 1/12	PP4 ≥ 0.70 0/12
F10	*PRELP*	●	○	○
	OMD	○	○	○
Metabolic	*IDH2*	●	○	○
F8	*NDUFA8*	○	○	○
	BMP10	●	○	○
F11	*SMAD9*	○	●	○
	SMAD6	●	○	○
	NTRK2	○	○	○
F15	*NTF3*	○	○	○
	FRS2	●	○	○
AF controls	*PITX2*	●	○	○
	KCNN3	●	○	○

SMAD9: atrial eGene+ · PP4-low

PITX2 / KCNN3: calibration controls also unsupported

B Coloc posterior and lead separation

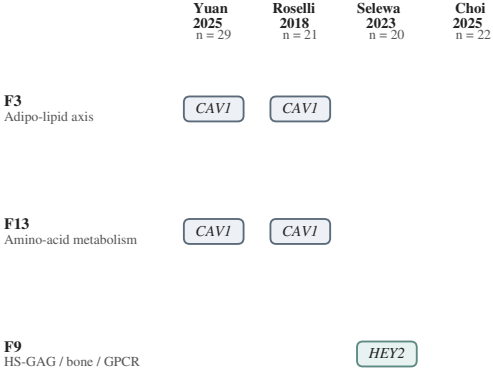
PP3-dominant 12/12 · PP4 support 0/12 · 0.46–1.92 Mb



Lead separation is descriptive and is not used in the coloc posterior.

C External AF-gene overlap

5 cells · 2 genes · 3/16 families · 0/64 FDR



Zero overlap: 13/16 families

Unique genes: CAV1 · HEY2

Labels = member overlap; no enrichment passed BH-FDR.

D Non-nested genetic traceability

No score · not Path A

Independent flags across 12 anchors

Family / role	Anchor	G 7/12	CORE E 1/12	C 0/12	Risk / prioritized 2/12	EXTERNAL Regulatory 2/12	Rare / functional 0/12
F10	*PRELP*	●	○	○	○	○	○
	OMD	○	○	○	○	○	○
Metabolic	*IDH2*	●	○	○	○	○	○
F8	*NDUFA8*	○	○	○	○	○	○
	BMP10	●	○	○	○	○	○
F11	*SMAD9*	○	●	○	○	○	○
	SMAD6	●	○	○	○	○	○
	NTRK2	○	○	○	○	○	○
F15	*NTF3*	○	○	○	○	○	○
	FRS2	●	○	○	○	○	○
AF controls	*PITX2*	●	○	○	●	●	○
	KCNN3	●	○	○	●	●	○

G = local GWAS · E = atrial eGene · C = PP4 ≥ 0.70

● present · ○ evaluated, not supported · — not evaluated